STA2002: Probability and Statistics II
Contingency Tables: Homogeneity

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Introduction

- In this lecture we will learn the basics of contingency tables. We first look at a method for testing whether two or more **multinomial distributions** are equal, sometimes called a **test for homogeneity**. Later, we would consider a **test for independence** of attributes of classification. Both of these lead to a similar test statistic. We would also demonstrate the flexibility of the chi-square test learned in class.

- **Suggested reading**: Chapter 9.2 of the textbook.

Homogeneity

- If we have two or more multinomial distributions, we would like to test whether these multinomial distributions are equal.
- We first consider the simple case, that we have two multinomial distributions, $j = 1, 2$.
- Meanwhile, there are k mutually exclusive and exhaustive categories $A_1, A_2, \ldots, A_k$.
- Probability p_{ij} are defined as,

$$p_{ij} = P(A_i; \text{under multinomial distribution } j),$$

where $i = 1, 2, \ldots, k$ and $j = 1, 2$.

Homogeneity

- In order to collect data, suppose that we conduct two independent experiments, $j = 1, 2$.
- Let the experiments be repeated independently for n_1 and n_2 times, respectively.
- let $Y_{11}, Y_{21}, \ldots, Y_{k1}$ be the frequencies of $A_1, A_2, \ldots, A_k$ associated with the n_1 independent trials of the first experiment.
- Similarly, let $Y_{12}, Y_{22}, \ldots, Y_{k2}$ be the frequencies of $A_1, A_2, \ldots, A_k$ associated with the n_2 independent trials of the second experiment.

Homogeneity

- Usually, the parameters p_{ij}, $i = 1, 2, \ldots, k$, $j = 1, 2$, are unknown.
- Suppose we wish to test the following hypotheses

$$H_0 : p_{11} = p_{12}, p_{21} = p_{22}, \ldots, p_{k1} = p_{k2}.$$

unknown parameter in polulation 1.

against,

$$H_1 : \text{ All alternative possibilities.}$$

- The meaning behind H_0 is that " the corresponding probabilities associated with the two independent multinomial distributions are equal."

Example 9.2-1

Example 1

To test two methods of instruction, 50 students are selected at random from each of two groups. At the end of the instruction period, each student is assigned a grade (A, B, C, D, or F) by an evaluating team. The data are recorded as follows:

	Grade					
	A	B	C	D	F	Totals
Group I	8	13	16	10	3	50
Group II	4	9	14	16	7	50

Example 9.2-1

- Question: whether there is a difference between the two methods of instruction?
- Let $p_{i,j}$ be the probability of final grade $i \in \{A, B, C, D, F\}$ for a randomly selected student in group $j \in \{1, 2\}$.
- Equivalently, we would like to test whether there is a difference in the grade distribution between group *I* and group *II*.
- The null hypothesis H_0 is proposed as,

$$H_0 : p_{i,1} = p_{i,2} = p_i, \quad i \in \{A, B, C, D, F\}.$$

- Question: How to make a decision?

Example 9.2-2

Example 2

Select, at random, 20 cars of each of two comparable major-brand models. All 40 cars are submitted to accelerated life testing; that is, they are driven many miles over very poor roads in a short time, and their failure times (in weeks) are recorded as follows:

Brand U:	25	31	20	42	39	19	35	36	44	26
	38	31	29	41	43	36	28	31	25	38
Brand V:	28	17	33	25	31	21	16	19	31	27
	23	19	25	22	29	32	24	20	34	26

Example 9.2-2

- Question: whether the two brands of cars have different distributions for the length of life under accelerated life testing? Whether one brand is significantly better than the other?
- If we use 23.5, 28.5, and 34.5 as dividing marks, then the data can be summarized as follows:

	A_1	A_2	A_3	A_4	Totals
Brand U	2	4	4	10	20
Brand V	8	6	6	0	20

- Question: How to make a decision?

Chi-square Test Statistic

- When n_j, $j = 1, 2$, is large, each of the following statistic

$$\sum_{i=1}^{k} \frac{(Y_{ij} - n_j p_{ij})^2}{n_j p_{ij}}, \quad j = 1, 2,$$

  has an approximate chi-square distribution with $k - 1$ degrees of freedom.
- Since the two experiments are independent, thus the two chi-square statistics are independent, then the sum,

$$\sum_{j=1}^{2} \sum_{i=1}^{k} \frac{(Y_{ij} - n_j p_{ij})^2}{n_j p_{ij}} \tag{1}$$

  is approximately chi-square with $k - 1 + k - 1 = 2k - 2$ degrees of freedom.

Chi-square Test Statistic

- Under H_0, the estimator of p_{i1} and p_{i2} is proposed as,

$$\hat{p}_i := \frac{Y_{i1} + Y_{i2}}{n_1 + n_2}, \quad i = 1, 2, \ldots, k-1$$

and for $i = k$, the estimator of $p_{k1} = p_{k2}$ is,

$$\hat{p}_k := 1 - \sum_{i=1}^{k-1} \frac{Y_{i1} + Y_{i2}}{n_1 + n_2}.$$

- Question: Think about the reason behind, and derive the mle of parameters in a multinomial distribution.

Chi-square Test Statistic

$Q = \sum_{j=1}^{2} \sum_{i=1}^{k} \frac{(Y_{ij} - n_j p_{ij})^2}{n_j p_{ij}} \sim \chi^2(n-1)$

- In Expression (1), replacing p_{ij} by $\hat{p}_i$ yields,

$$Q := \sum_{j=1}^{2} \sum_{i=1}^{k} \frac{(Y_{ij} - n_j \hat{p}_i)^2}{n_j \hat{p}_i}. \tag{2}$$

↳ derived under H_0

- Under H_0, the statistic Q has an approximate chi-square distribution with $(2k-2)-(k-1) = k-1$ degrees of freedom.

$p_{i1} = p_{i2}$, $i = 1, 2 \ldots k$

- Pay attention: Under $H_0 \cup H_1$, there $2k-2$ independent and unknown parameters. Under H_0, there are $k-1$ independent and unknown parameters. The difference between the number of parameters under $H_0 \cup H_1$ and under H_0 is $(2k-2)-(k-1) = k-1$.

Decision Rule

- Once collected the data, q denotes the realized value of test statistic Q.
- Reject H_0 at significance level α, if

$$q \geq \chi^2_\alpha(k-1).$$

- The critical region is of the form,

$$C := \{Q \geq \chi^2_\alpha(k-1)\}.$$

- The p-value is,

$$\text{p-value} = P(Q \geq q; H_0) \simeq P(\chi^2(k-1) \geq q).$$

Example 9.2-1 Again

- Solution.
  Under H_0, The estimation of p_{i1} and p_{i2} is,

$$\hat{p}_A = \frac{8+4}{100} = 0.12, \quad \hat{p}_B = \frac{13+9}{100} = 0.22,$$

  and,

$$\hat{p}_C = 0.30, \quad \hat{p}_D = 0.26, \quad \hat{p}_F = 0.10.$$

- The realized value q of Q is,

$$\begin{aligned} q &= \frac{(8-6)^2}{6} + \frac{(13-11)^2}{11} + \frac{(16-15)^2}{15} + \frac{(10-13)^2}{13} + \frac{(3-5)^2}{5} \\ &\quad + \frac{(4-6)^2}{6} + \frac{(9-11)^2}{11} + \frac{(14-15)^2}{15} + \frac{(16-13)^2}{13} + \frac{(7-5)^2}{5} \\ &= \frac{4}{6} + \frac{4}{11} + \frac{1}{15} + \frac{9}{13} + \frac{4}{5} + \frac{4}{6} + \frac{4}{11} + \frac{1}{15} + \frac{9}{13} + \frac{4}{5} = 5.18. \end{aligned}$$

Example 9.2-1 Again

- Compare q with $\chi^2_\alpha(4)$. When $\alpha = 5\%$, $\chi^2_\alpha(4) = 9.488$, and

$$q = 5.18 < \chi^2_\alpha(4) = 9.488.$$

- p-value is 0.269. p-value $> \alpha = 5\%$.
- Decision: Retain H_0 at significance level $\alpha = 5\%$.
- Interpretation: Based on the available and observed data, we cannot say that there is a difference between the two methods of instruction. □

Example 9.2-2 Again

- Solution.
  Under H_0, The estimation of p_{i1} and p_{i2} is,

$$\hat{p}_1 = 1/4,\ \hat{p}_2 = 1/4,\ \hat{p}_3 = 1/4, \hat{p}_4 = 1/4.$$

- The realized value of Q is,

$$\begin{aligned} q &= \frac{(2-5)^2}{5} + \frac{(4-5)^2}{5} + \frac{(4-5)^2}{5} + \frac{(10-5)^2}{5} + \frac{(8-5)^2}{5} \\ &\quad + \frac{(6-5)^2}{5} + \frac{(6-5)^2}{5} + \frac{(0-5)^2}{5} \\ &= \frac{72}{5} = 14.4 > 7.815 = \chi^2_{0.05}(3). \end{aligned}$$

Example 9.2-2 Again

- Compare q with $\chi^2_\alpha(3)$. When $\alpha = 5\%$, $\chi^2_\alpha(3) = 7.815$, and

$$q = 14.4 > \chi^2_{0.05}(3) = 7.815.$$

- p-value is 0.0024. p-value $< \alpha = 5\%$.
- Decision: Reject H_0 at significance level $\alpha = 5\%$. As p-value is small, we would reject H_0 even at significance level 1%.
- Interpretation: Based on the available and observed data, statistically, the two brands of cars have significant different distributions for the length of life under accelerated life testing. Brand U seems better than brand V. □